

A negative answer to the cardinal condensation-rank problem for $\ell_\infty(\Gamma)$

Packet for arXiv:1104.4896

Source problem

Gazor's paper *Condensation rank of injective Banach spaces* (arXiv:1104.4896) defines the usual condensation derivative and proves a lower-bound theorem for infinite-dimensional injective Banach spaces. At the end of Section 3 the author asks the following cardinal analogue: after defining a κ -condensation derivative, if $\kappa = 2^{|\Gamma|}$, is the κ -condensation rank of $\ell_\infty(\Gamma)$ the first ordinal of cardinality κ ?

The natural nontrivial convention is that a point is a κ -condensation point of A when every neighbourhood of the point meets A in at least κ points. Under this convention, the answer is no: the rank is one cardinal step larger.

Definitions

Let κ be an infinite cardinal and X a topological space. For $A \subseteq X$, put

$$D_\kappa(A) = \{x \in X : |U \cap A| \geq \kappa \text{ for every neighbourhood } U \text{ of } x\}.$$

The transfinite iterates are defined by

$$D_\kappa^0(A) = A, \quad D_\kappa^{\alpha+1}(A) = D_\kappa(D_\kappa^\alpha(A)), \quad D_\kappa^\lambda(A) = \bigcap_{\alpha < \lambda} D_\kappa^\alpha(A)$$

for limit ordinals λ . Let $\rho_\kappa(A)$ be the least ordinal α such that $D_\kappa^\alpha(A) = D_\kappa^{\alpha+1}(A)$, and set

$$\text{CR}_\kappa(X) = \sup\{\rho_\kappa(A) : A \subseteq X\}.$$

For a cardinal μ , let ω_μ denote the initial ordinal of cardinality μ . Thus the ordinal asked for in the source problem is ω_κ , while $\omega_{\kappa+}$ is the first ordinal whose cardinality is strictly larger than κ .

Proof intuition

The upper bound is purely cardinal: along a strictly decreasing derivative chain one can choose a different point lost at each successor step, so a space of cardinality κ cannot support κ^+ many strict losses.

The lower bound is a transfinite rank-tree construction inside $\ell_\infty(\Gamma)$. The cube $\{0,1\}^\Gamma$ gives $\kappa = 2^{|\Gamma|}$ many pairwise distance-one directions. These directions let us place κ many small copies of any previously built rank block near a root, in countably many shrinking shells. The root then becomes a κ -condensation point of every earlier derivative, survives one derivative longer than the copies, and disappears immediately after that. Limit ordinals are handled by separated sums of cofinally many earlier rank blocks.

Main theorem

Theorem 1. *Let Γ be an infinite set and let $\kappa = 2^{|\Gamma|}$. With the convention above,*

$$\text{CR}_\kappa(\ell_\infty(\Gamma)) = \omega_{\kappa+}.$$

Consequently the equality proposed in the final problem of arXiv:1104.4896, $\text{CR}_\kappa(\ell_\infty(\Gamma)) = \omega_\kappa$, is false.

Proof. First note that

$$|\ell_\infty(\Gamma)| = \kappa.$$

Indeed, $\{0, 1\}^\Gamma \subseteq \ell_\infty(\Gamma)$ has cardinal $2^{|\Gamma|}$, while the set of all scalar functions on Γ has cardinal

$$(2^{\aleph_0})^{|\Gamma|} = 2^{|\Gamma|}$$

because Γ is infinite.

Upper bound. Let $A \subseteq \ell_\infty(\Gamma)$. If $D_\kappa^\alpha(A) \supsetneq D_\kappa^{\alpha+1}(A)$ for every $\alpha < \lambda$, then we may choose

$$x_\alpha \in D_\kappa^\alpha(A) \setminus D_\kappa^{\alpha+1}(A) \quad (\alpha < \lambda).$$

These points are distinct: if $\alpha < \beta$, then

$$x_\beta \in D_\kappa^\beta(A) \subseteq D_\kappa^{\alpha+1}(A),$$

whereas $x_\alpha \notin D_\kappa^{\alpha+1}(A)$. Hence $|\lambda| \leq |A| \leq \kappa$. Therefore no derivative chain in $\ell_\infty(\Gamma)$ is strictly decreasing through all ordinals below ω_{κ^+} , and

$$\text{CR}_\kappa(\ell_\infty(\Gamma)) \leq \omega_{\kappa^+}.$$

Two elementary placement facts. Let

$$V = \{0, 1\}^\Gamma \setminus \{0\} \subseteq \ell_\infty(\Gamma).$$

Then $|V| = \kappa$, $\|v\| = 1$ for every $v \in V$, and distinct points of V have distance 1.

We shall use two standard consequences of this separated family.

First, if $\{B_i : i \in I\}$ is a family of closed subsets of $\ell_\infty(\Gamma)$, $|I| \leq \kappa$, and each B_i has cardinality at most κ , then scaled translates of the B_i 's can be placed in pairwise uniformly separated balls inside any prescribed nonempty ball. For such a uniformly separated union,

$$D_\kappa^\alpha\left(\bigcup_i B_i\right) = \bigcup_i D_\kappa^\alpha(B_i) \quad \text{for every ordinal } \alpha. \quad (1)$$

The reason is local: every point has a neighbourhood meeting at most one component.

Second, suppose $\{B_i : i \in I\}$ is a family of closed sets, indexed by

$$I = \mathbb{N} \times \kappa,$$

and suppose each B_i has cardinality at most κ . Given a point p and a radius $r > 0$, choose numbers $a_n \downarrow 0$ so fast that the shells remain separated, for instance $a_n = r2^{-2n-4}$. Put the components in balls of radius $a_n/100$ around points $p + a_n v_\xi$, where $\xi < \kappa$ and $v_\xi \in V$. The resulting closed set

$$F = \{p\} \cup \bigcup_{(n,\xi) \in I} B_{n,\xi}$$

has only one possible cross-component κ -condensation point, namely p . Away from p , a small neighbourhood meets only finitely many shells and, in each such shell, at most one component.

Every neighbourhood of p , however, contains all components in sufficiently small shells and therefore contains κ many components.

Rank blocks. We prove by transfinite induction that for every nonzero $\alpha < \omega_{\kappa^+}$ and every nonempty ball B in $\ell_\infty(\Gamma)$, there is a closed set $S_\alpha \subseteq B$, with $|S_\alpha| \leq \kappa$, such that

$$D_\kappa^\beta(S_\alpha) \neq \emptyset \quad (\beta < \alpha), \quad D_\kappa^\alpha(S_\alpha) = \emptyset. \quad (2)$$

Thus S_α has κ -condensation height exactly α .

For $\alpha = 1$, take a singleton.

Assume first that α is a nonzero limit ordinal. Since $\alpha < \omega_{\kappa^+}$, $|\alpha| \leq \kappa$. Choose a cofinal set $C \subseteq \alpha$ with $|C| \leq \kappa$. By the induction hypothesis, for each $\gamma \in C$ choose a closed rank block of height γ , and place these blocks as a uniformly separated family inside B . Equation (1) shows that the derivative of the union is the union of the derivatives. Given $\beta < \alpha$, some $\gamma \in C$ satisfies $\beta < \gamma$, so the β -th derivative of the union is nonempty. At stage α , every component has already disappeared. Hence (2) holds.

Now suppose $\alpha = \beta + 1$. If $\beta = 0$, this is again the singleton case. If $\beta > 0$, choose for every $(n, \xi) \in \mathbb{N} \times \kappa$ a closed block $B_{n, \xi}$ of height β , scaled small enough to fit in the fan construction above, and set

$$S_{\beta+1} = \{p\} \cup \bigcup_{(n, \xi) \in \mathbb{N} \times \kappa} B_{n, \xi}.$$

For every $\eta < \beta$, each component has nonempty D_κ^η , and every neighbourhood of p contains κ many components. Hence $p \in D_\kappa^{\eta+1}(S_{\beta+1})$, and consequently $p \in D_\kappa^\beta(S_{\beta+1})$. The components have empty β -th derivative, so the fan observation gives

$$D_\kappa^\beta(S_{\beta+1}) = \{p\}.$$

The singleton $\{p\}$ has empty κ -condensation derivative, so

$$D_\kappa^{\beta+1}(S_{\beta+1}) = \emptyset.$$

Moreover the same argument gives nonemptiness at every earlier stage. This completes the induction.

Since rank blocks exist for every nonzero $\alpha < \omega_{\kappa^+}$, the supremum of the ranks of subsets of $\ell_\infty(\Gamma)$ is at least ω_{κ^+} . Together with the upper bound, this proves

$$\text{CR}_\kappa(\ell_\infty(\Gamma)) = \omega_{\kappa^+}.$$

□

Remarks on conventions

The source problem is meaningful only under an “at least κ ” interpretation of κ -condensation. If one instead required strictly more than κ points in every neighbourhood while also taking $\kappa = 2^{|\Gamma|} = |\ell_\infty(\Gamma)|$, the first derivative would be empty for every subset, so the proposed equality would fail trivially. The theorem above addresses the nontrivial convention.

Novelty and verification

The local index check found no previous run packet for arXiv:1104.4896. A bounded web search for the phrases “aleph-condensation”, “condensation rank”, and “ $\ell_\infty(\Gamma)$ ” found only the source arXiv paper, not a later answer to the final problem. The proof uses only elementary cardinal arithmetic, transfinite induction, and the explicit separated family $\{0, 1\}^\Gamma \subseteq \ell_\infty(\Gamma)$.

Reference

M. Gazor, *Condensation rank of injective Banach spaces*, arXiv:1104.4896, 2011.